STAGE P2 · SPECIFICATION & PROTOTYPE

VELARIS.

System Architecture, Relational Data Model, Critical Transaction Sequences & Live Working Prototype.

FRONTEND & UX LEAD
Predut Andreea-Denisa
Catalog, Cart, Checkout & Responsive UI

BACKEND & SYSTEMS LEAD
Sapca Roman-Stefan
APIs, EF Core, Concurrency & Payments

P2 MILESTONE

Specification Frozen for Stage P3

Commit Tag: **P2** | SQLite + EF Core

DOMAIN FOUNDATION

Catalogue Scope & Product Niche

CATEGORY 1

Heavyweight Hoodies

8 SKUs of ultra-dense 450 GSM organic cotton french terry with pre-shrunk boxy silhouette.

SKUs: VEL-HOD-01 to 08
Price: 340 – 420 RON
Sizes: S, M, L, XL

CATEGORY 2

Boxy & Relaxed Tees

8 SKUs of combed 280 GSM cotton featuring double-needle stitched collars and drop shoulders.

SKUs: VEL-TEE-01 to 08
Price: 160 – 210 RON
Sizes: XS, S, M, L, XL

CATEGORY 3

Structured Denim & Cargo

8 SKUs of 14oz Japanese raw selvedge denim and heavyweight double-knee utility twill trousers.

SKUs: VEL-DNM-01 to 08
Price: 380 – 480 RON
Waist: 30, 32, 34, 36

CATEGORY 4

Weatherproof Outerwear

8 SKUs of structured wool overcoats, technical ripstop shell jackets and quilted liners.

SKUs: VEL-OUT-01 to 08
Price: 520 – 780 RON
Sizes: S, M, L

Requirements Implementation (A1–A6)

CODE	REQUIREMENT	TECHNICAL IMPLEMENTATION (1–2 SENTENCES)	OWNER
A1	**Product Catalog**	Paginated grid capped at 12 garments per page (below 24 limit), dedicated category views, and responsive detail views with fabric GSM density.	Andreea
A2	**Basic Search**	Text search matching garment names and fabric compositions with clear contextual empty-state messaging when no products are matched.	Andreea
A3	**Shopping Cart**	Cart state recalculation on the server, immediate rejection of quantities exceeding stock, and persistent database storage for logged-in users.	Andreea
A4	**User Accounts**	Registration and login secured via ASP.NET Core Identity with PBKDF2/Argon2 password hashing and saved shipping address profiles.	Stefan
A5	**Checkout Flow**	Multi-step sequence: address confirmation, 2 distinct delivery carrier options (Standard 15 RON vs. Express 30 RON), summary, and Stripe redirect.	Andreea
A6	**Sandbox Payment**	Integration with Stripe Test Mode; orders become *Paid* strictly upon webhook confirmation (`checkout.session.completed`), not browser redirect.	Stefan

MANDATORY SCOPE

Requirements Implementation (A7–A12)

CODE	REQUIREMENT	TECHNICAL IMPLEMENTATION (1–2 SENTENCES)	OWNER
A7	**Stock Management**	Atomic inventory decrement upon confirmed payment webhook; optimistic concurrency tokens (`RowVersion`) prevent overselling last unit.	Stefan
A8	**Order History**	Customer dashboard displaying all past orders with date, line items, unit prices, total cost, and status badges (New, Paid, Shipped, Delivered).	Andreea
A9	**Admin Panel**	Custom back-office for product and category CRUD with image file upload, and order status lifecycle transition management.	Stefan
A10	**Basic Security**	Server-side model validation, automatic anti-CSRF token verification on all POST actions, ORM parameterization, and `[Authorize(Roles="Admin")]`.	Stefan
A11	**Responsive Design**	Mobile-first Tailwind CSS layout tested down to 360px viewport width (tested on iPhone SE and small Android viewports).	Andreea
A12	**Automated Tests**	Over 10 xUnit integration tests covering cart calculations, stock decrement bounds, and webhook receipt, executable with `dotnet test`.	Stefan

BONUS FEATURES

Approved Bonus Features Selection

BONUS B1 — 5 Points

Filtered Search

Faceted filtering combining categories, price range sliders, and in-stock availability toggles, with sorting by price and date.

Technical Proof:
`All filter parameters are serialized into clean URL queries: ?cat=hoodies&min=200&sort=price-asc, preserving browser history.`

BONUS B5 — 3 Points

Discount Codes

Server-validated promotional codes supporting percentage (e.g. 10%) or fixed amounts, with expiration dates and maximum redemption limits.

Technical Proof:
`Coupon deduction is validated entirely on the backend; client cannot tamper with the final captured Stripe session amount.`

BONUS B7 — 2 Points

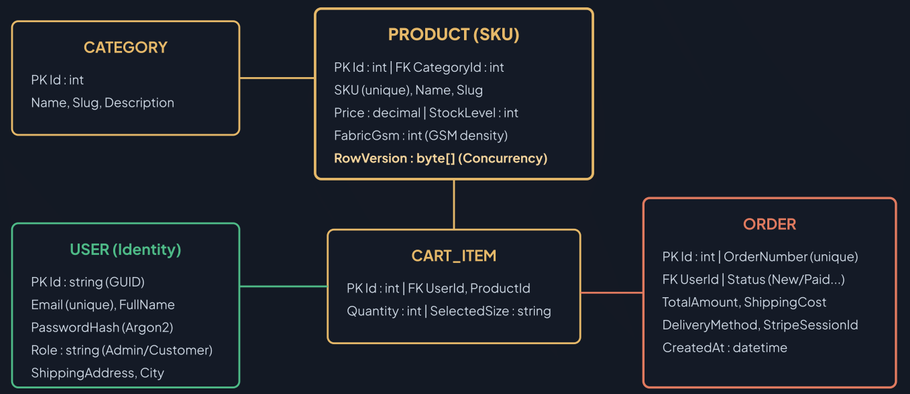
Customer Wishlist

Authenticated favorites board where customers save capsule staples, with a single-click action to move items directly into the shopping cart.

Technical Proof:
`Stored in SQLite table WishlistItems with composite key (UserId, ProductId) ensuring zero duplicate entries.`

RELATIONAL SCHEMA

Entity-Relationship Data Model (ERD)



Layered System Component Architecture

PRESENTATION LAYER

Storefront & Admin

- ASP.NET Core Razor Pages & ViewComponents
- Tailwind CSS Design Tokens (Onyx & Gold)
- Responsive mobile layouts down to 360px (A11)
- Custom back-office at `/Admin` with product CRUD

APPLICATION CORE

ASP.NET Core Services

- Anti-CSRF validation middleware & Cookie Auth
- Catalog & full-text search engine (A1, A2)
- Persistent cart state manager (A3)
- Stripe Webhook handler with signature verification
- Optimistic concurrency lock manager (A7)

PERSISTENCE & SANDBOX

Data & External APIs

- SQLite Database (`velaris.db`) via EF Core ORM
- Automatic 32-SKU database migration seed
- Stripe Test Mode API & local CLI listener
- Mailtrap SMTP testing service for order receipts

ENDPOINTS CONTRACT

RESTful API Endpoints & Access Levels

METHOD	ROUTE	ACCESS LEVEL	OPERATION & REQUIREMENT
POST	/api/auth/register	Public	Registers customer account with PBKDF2/Argon2 hashing (A4).
POST	/api/auth/login	Public	Authenticates user and sets secure HTTP-only session cookie (A4).
GET	/api/catalog/products	Public	Paged product query with category, search and sort params (A1, A2, B1).
POST	/api/cart/items	Public / Auth	Adds item line to cart with server-side stock validation (A3).
POST	/api/checkout/create-session	Auth Customer	Validates delivery method and initiates Stripe Checkout Session (A5).
POST	/api/stripe/webhook	Stripe Service	Verifies signature, decrements stock atomically, sets status *Paid* (A6, A7).
GET/POST	/api/admin/products/*	Admin Only	Garment CRUD with image uploads; customers blocked with 403 Forbidden (A9, A10).

TRANSACTION FLOW

Critical Flow: Checkout → Stripe → Webhook → Stock

Step-by-Step Critical Path (A6, A7)

1. **Checkout Submission:** Customer reviews address and selects delivery courier (Fan Courier 15 RON vs. DPD Express 30 RON).
2. **Session Creation:** App verifies stock and initializes a Stripe Checkout Session with order metadata.
3. **Card Processing:** Customer enters test card (4242...) in Stripe sandbox.
4. **Async Webhook Delivery:** Stripe sends cryptographically signed POST to `/api/stripe/webhook`.
5. **Atomic Concurrency Decrement:** Transaction locks garment rows; stock decreases, `RowVersion` updates.
6. **Order State Transition:** Status updates: *New* → *Paid*. Confirmation page displays final invoice number.

Race Condition Mitigation

What happens when two shoppers purchase the last available Hoodie unit simultaneously?

```
// EF Core Optimistic Concurrency try { product.StockLevel -= item.Quantity; await
_db.SaveChangesAsync(); // Checks RowVersion order.Status = OrderStatus.Paid; } catch
(DbUpdateConcurrencyException) { // Transaction safely rolled back order.Status = OrderStatus.Unpaid;
await RefundStripePaymentAsync(session.PaymentIntentId); }
```

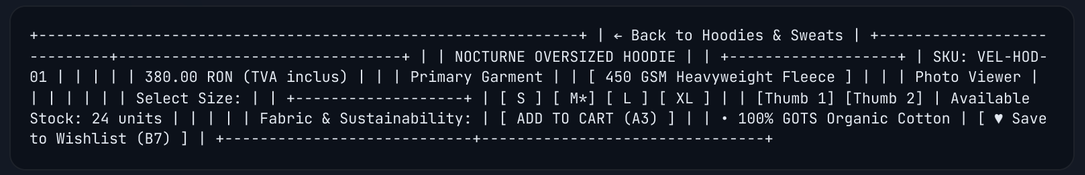
Guarantees inventory never drops into negative stock and orders cannot be fulfilled without reserved stock.

Wireframes: Storefront & Product Detail

Wireframe 1: Product Catalog (A1, A2, B1)



Wireframe 2: Product Detail (A1, B7)



USER INTERFACE

Wireframes: Cart, Checkout & Administration

Wireframe 3 & 4: Cart & Checkout (A3, A5)

```
+-------------------------------------------------------------+ | SHOPPING CART & CHECKOUT WIZARD | +-------------
-------------------------------------------------+ | 1. Nocturne Hoodie (M) x1 .............. 380.00 RON | | 2. Boxy
Heavyweight Tee (L) x2 ......... 360.00 RON | | Promo Code (B5): [ VELARIS10 ] → Discount: -74.00 RON | +----------
-------------------------------------------------+ | Select Delivery Carrier (A5 - at least 2 options): | | (o)
Fan Courier Standard (2-3 business days) ..... 15.00 RON| | ( ) DPD Express Priority (Next business day) .....
30.00 RON| +-------------------------------------------------------------+ | Total to Pay: 681.00 RON (Server-Side
Recalculated) | | [ PROCEED TO STRIPE SANDBOX (A6) → ] | +-------------------------------------------------------------
----+
```

Wireframe 5: Admin Panel (A9, A10)

```
+-------------------------------------------------------------+ | VELARIS ADMIN DASHBOARD Logged: admin@velaris | |
[ Products (32) ] [ Categories (4) ] [ Orders (14) ] | +-------------------------------------------------------------
--+ | Filter Orders: [ Status: Paid ↓ ] | | Order #VEL-8941 | Andreea Predut | 681 RON | [ Mark Shipped]| | Order
#VEL-8940 | Roman Sapca | 380 RON | [ Mark Shipped]| +-------------------------------------------------------------
+ | Product CRUD Action: | | [ + Add Garment SKU ] Upload Image: [ Choose file... ] | | Inputs: Name, Category,
GSM, Price, Initial Stock Level | +-------------------------------------------------------------+
```

LIVE VERIFICATION

Working Prototype Demonstration (P2 Demo)

DEMO STEP 1

Live Database Seed

Demonstrating zero-config SQLite startup. 32 distinct clothing SKUs loaded across 4 categories with real Unsplash images and GSM fabric specs.

```
Command: dotnet run
URL: http://localhost:5000
Auto-migration & seed verified.
```

DEMO STEP 2

Interactive Catalog

Navigating paginated products (12/page), category filtering, search query execution, and mobile viewport simulation down to 360px.

```
Features shown:
• Category tabs
• Empty state search feedback
• Responsive 360px grid
```

DEMO STEP 3

User Auth & Role Guard

Registering a customer account with PBKDF2/Argon2 password hashing, session login, and verifying that customers cannot access `/Admin`.

```
Credentials for testing:
User: customer@velaris.com
Admin: admin@velaris.com
```

EXECUTION MILESTONES

Roadmap to Presentation 3 (Final Delivery)

SPRINT 4 (WEEKS 8-9)

Cart & Order History

- A3: Shopping cart state persistence
- A8: Customer past orders portal
- A9: Admin product & category CRUD
- A10: Server-side route authorization

SPRINT 5 (WEEKS 10-11)

Payments & Testing

- A5: Multi-step checkout wizard
- A6: Stripe sandbox webhook listener
- A7: Concurrency lock (RowVersion)
- A12: 10+ automated tests in 1 command

SPRINT 6 (WEEK 12)

Bonuses & P3 Defense

- B1: URL query filtered search
- B5: Server discount coupons engine
- B7: Customer wishlist & favorites
- Live scenario trial run & P3 delivery

VELARIS.

P2 Technical Specification & Prototype Complete

Predut Andreea-Denisa & **Sapca Roman-Stefan**

GitHub Repository: `https://github.com/Denica2208/VELARIS`

Milestone Commit Tag: **P2**

Questions & Live Prototype Demonstration